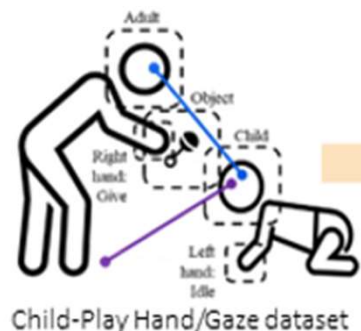


Understand and Predict Human Social Signals in videos

Hong Nguyen, SAIL USC ECE



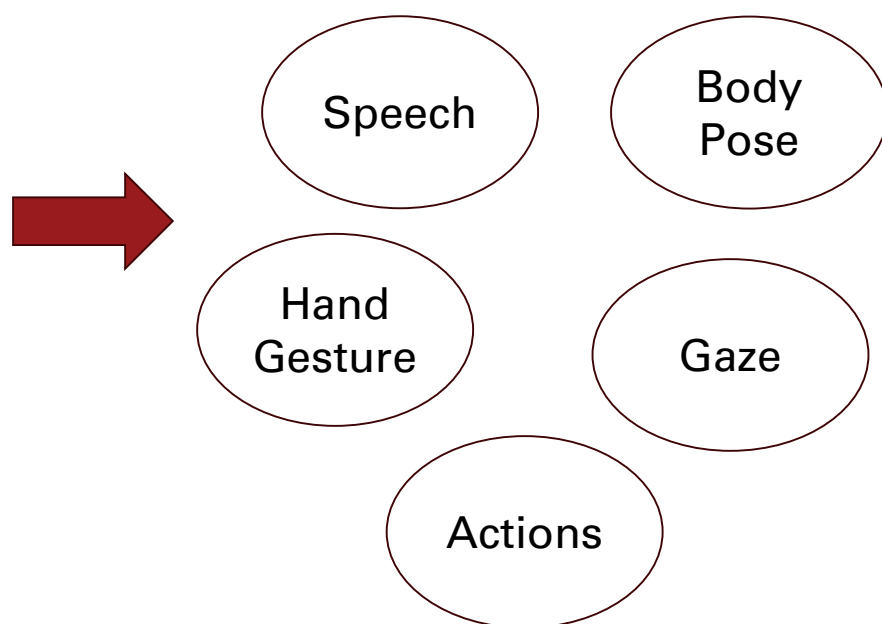
Introduction



Child-Adult



Police-Civilian



Hypothesis

- Speech and Gesture are observation of a same thought
- Gestures are highly sensitive to temporal dynamics in video

Encode Gestures in Videos?

1. Dual-Stream Architecture (Spatial-Temporal Separation)

- This explicit separation enables the network to jointly capture *appearance* and *motion dynamics* without overfitting to static spatial cues. (Figure 2)

2. Motion Tokenization and Temporal Attention Mechanism

- Optical flow features are transformed into motion tokens, encoding frame-to-frame movement information.

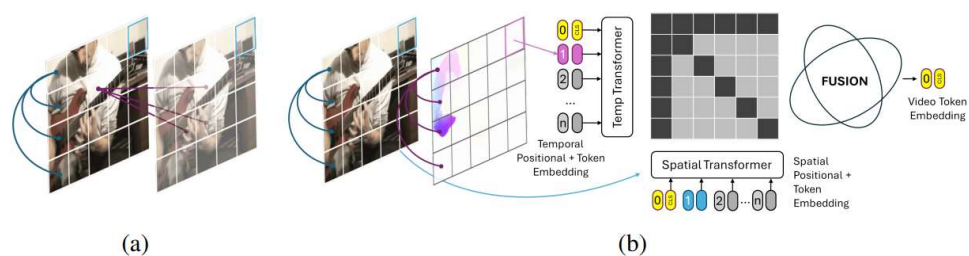
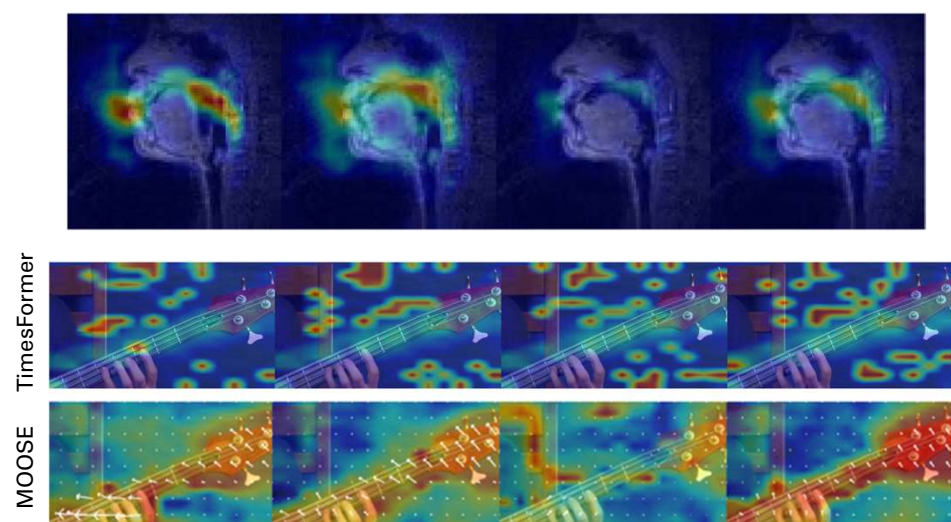


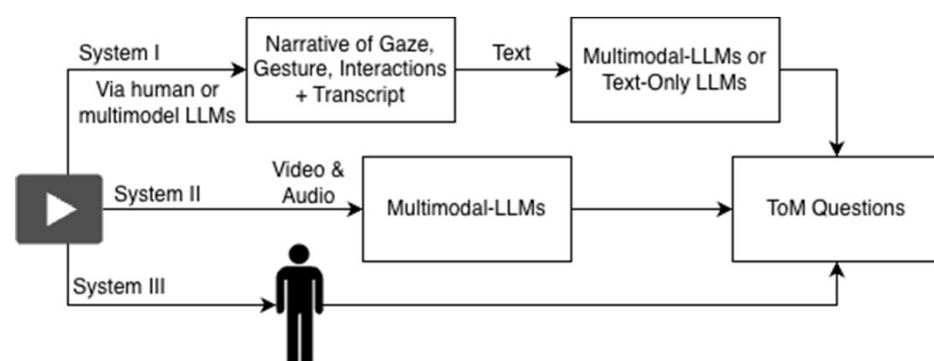
Figure 1: (a) Conventional attention-based methods perceive temporal dynamics with cross-spatial attention and (b) Our approach considers temporal dynamics as a series of vector fields (via optical flow). Flow vectors is grouped in patches to match image patches

4. Integration of Pre-trained Backbones

5. Temporal Aggregation and Video-Level Representation



Social Interaction Reasoning

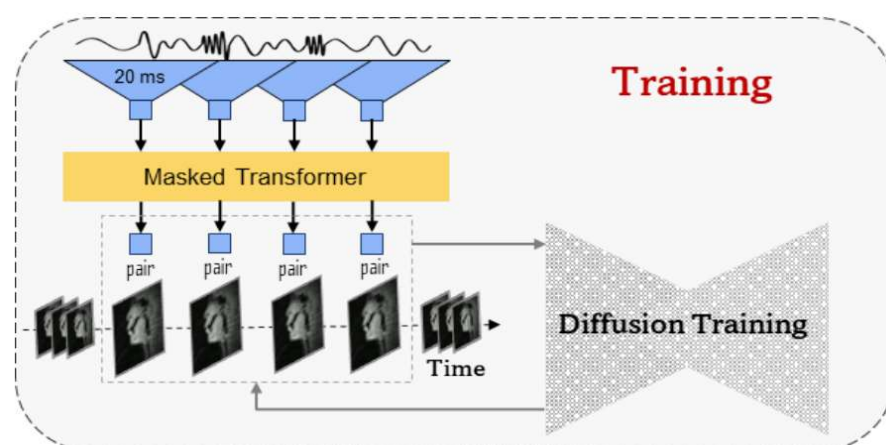


Research Questions:

Previous results suggest that M-LLMs underutilize audiovisual cues

- Could converting audiovisual cues to text-based input improve performance?
- How does M-LLM performance compare to human performance?
- How similar is the underlying reasoning between M-LLMs and human?

From Articulator Gestures to Speech... and Reverse



- Objective: Synthesize articulator movement of any speech via rt-MRI
- Designed Speech-Conditional **Diffusion model**